

```
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)  
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas)  
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas)  
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas)  
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas)  
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil)
```

```
import pandas as pd  
import numpy as np
```

```
print(pd.__version__)
```

```
2.2.2
```

series=Creation of 1D array in column wise with index

```
#from list  
s=pd.Series([1,2,3,4,5])  
print(s)
```

```
0    1  
1    2  
2    3  
3    4  
4    5  
dtype: int64
```

Code cell output actions

```
#custom index  
s=pd.Series([1,2,3,4,5],index=['a','b','c','d','e'])  
print(s)
```

```
a    1  
b    2  
c    3  
d    4  
e    5  
dtype: int64
```

```
#dictionary  
students={  
    "name":"vamshi",  
    "branch":"aiml"  
}  
ser=pd.Series(students)  
print(ser)
```

```
name      vamshi  
branch    aiml  
dtype: object
```

```
#arithmetic operations  
s=pd.Series([10,20,30,40,50])  
s*2
```

	θ
0	20
1	40
2	60
3	80
4	100

dtype: int64

s+2

	θ
0	12
1	22
2	32
3	42
4	52

dtype: int64

s.describe()

Code cell output actions

	θ
count	5.000000
mean	30.000000
std	15.811388
min	10.000000
25%	20.000000
50%	30.000000
75%	40.000000
max	50.000000

dtype: float64

s[s>30]

	θ
3	40
4	50

dtype: int64

```
#vectorised string operations
f=pd.Series(['vamsi','shaik','rani'])
f.str.lower()
```

```

      0
0  vamsi
1  shaik
2   rani
```

dtype: object

```
f.str.len()
```

```

      0
0     5
1     5
2     4
```

dtype: int64

```
students={
    "name":["vamsi","shaik","rani"],
    "aiml":["aiml","aiml","aiml"]
}
df=pd.DataFrame(students)
print(df)
```

```

   name branch
0  vamsi   aiml
1  shaik   aiml
2   rani   aiml
```

```
std=[
    {"name":"shaik","age":20,"branch":"aiml"},
    {"name":"vamsi","age":20,"branch":"aiml"},
    {"name":"rani","age":20,"branch":"aiml"}
]
s=pd.DataFrame(std)
print(s)
```

```

   name  age branch
0  shaik   20   aiml
1  vamsi   20   aiml
2   rani   20   aiml
```

```
#from numpy array
arr=np.random.rand(3,4)
c=pd.DataFrame(arr,columns=['a','b','c','d'],index=['x','y','z'])
print(c)
```

```

      a      b      c      d
x  0.707642  0.858774  0.488122  0.236267
y  0.507720  0.420738  0.371339  0.543168
z  0.782599  0.874954  0.482511  0.887777
```

```
data={
    'Name':['ravi','hari','siri','raju'],
    'Age':[28,25,22,28],
    'Score':[88,75,92,81],
    'Grade':['A','B','A','A']
}
d=pd.DataFrame(data)
print(d)
```

	Name	Age	Score	Grade
0	ravi	28	88	A
1	hari	25	75	B
2	siri	22	92	A
3	raju	28	81	A

```
d.head()
```

	Name	Age	Score	Grade
0	ravi	28	88	A
1	hari	25	75	B
2	siri	22	92	A
3	raju	28	81	A

Next steps:

[Generate code with d](#)[New interactive sheet](#)

```
d.tail()
```

	Name	Age	Score	Grade
0	ravi	28	88	A
1	hari	25	75	B
2	siri	22	92	A
3	raju	28	81	A

```
d.columns
```

```
Index(['Name', 'Age', 'Score', 'Grade'], dtype='object')
```

```
d.index
```

```
RangeIndex(start=0, stop=4, step=1)
```

```
d.dtypes
```

Name	object
Age	int64
Score	int64
Grade	object

```
dtype: object
```

```
d.unique
```

```
pandas.core.frame.DataFrame.nunique
def nunique(axis: Axis=0, dropna: bool=True) -> Series
```

</usr/local/lib/python3.12/dist-packages/pandas/core/frame.py>
Count number of distinct elements in specified axis.

Return Series with number of distinct elements. Can ignore NaN values.

```
d['Name']
```

	Name
0	ravi
1	hari
2	siri
3	raju

dtype: object

```
d.Name
```

	Name
0	ravi
1	hari

Code cell output actions

3 raj

dtype: object

```
d[['Name', 'Score']]
```

	Name	Score
0	ravi	88
1	hari	75
2	siri	92
3	raju	81

```
d.loc[0]
```

	0
Name	ravi
Age	28
Score	88
Grade	A

dtype: object

```
d.loc[0:2]
```

	Name	Age	Score	Grade
0	ravi	28	88	A
1	hari	25	75	B
2	siri	22	92	A

d.loc[0:2, 'Name': 'Score']				
	Name	Age	Score	
0	ravi	28	88	
1	hari	25	75	
2	siri	22	92	

d.iloc[0]				
	0			
Name	ravi			
Age	28			
Score	88			
Grade	A			
dtype: object				

Code cell output actions				
d.iloc[0:2]				
	Name	Age	Score	Grade
0	ravi	28	88	A
1	hari	25	75	B

d.iloc[0:2,0:2]				
	Name	Age		
0	ravi	28		
1	hari	25		

d.iloc[-1]				
	3			
Name	raju			
Age	28			
Score	81			
Grade	A			
dtype: object				

d.loc[0, 'Name']				
'ravi'				

```
d.iloc[0,0]
```

```
'ravi'
```

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Code cell output actions